

Problem Set 10: AI-Econ Nexus

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [date — before finals week]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with $\langle AI \rangle$ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (What is a replication worth? Bayes on silicon evidence — 15 points). *You hold a prior of $P(F) = 0.3$ that your agent population is a faithful simulator of the target human population (versus $\neg F$: it produces answers by recitation and surface patterns without a faithful behavioral process).*

- (a) **(5 pts)** *You run a famous experiment and the agents reproduce the published result. Because the result saturates the training data, recitation predicts this success almost as well as fidelity: $P(\text{reproduce} \mid F) = 0.9$ and $P(\text{reproduce} \mid \neg F) = 0.85$. Compute the posterior $P(F \mid \text{reproduce})$ and state, in one sentence, what the small movement teaches.*
- (b) **(5 pts)** *Starting from your posterior in (a), you now test a novel, preregistered, out-of-corpus prediction (a design variation with no published answer): $P(\text{match} \mid F) = 0.7$, $P(\text{match} \mid \neg F) = 0.2$. Compute the posterior after a successful match. Why is this test so much more informative — answer in terms of the likelihood ratio.*
- (c) **(3 pts)** *Your labmate proposes boosting credibility by replicating THREE more famous results. Explain in two sentences why the posterior gain is small even after all three, and what single study design would move it more.*
- (d) **(2 pts)** *One sentence: what does this calculation imply about how your capstone’s simulation section should be structured?*

Question 2 (Reading a stability report — 15 points). *Your audit battery (Lab 10) runs a dictator-game pilot across two models and three prompt paraphrases (same content, different wording), at fixed temperature. Mean percent given:*

	Paraphrase 1	Paraphrase 2	Paraphrase 3
Model A	27	29	41
Model B	33	34	36

For reference, human dictator-game means vary across comparable lab replications by roughly ± 5 percentage points around a common mean — treat spreads wider than about 10 points as clearly exceeding the band.

- (a) **(4 pts)** Compute each model’s paraphrase range (the prompt-sensitivity index) and the cross-model gap at each paraphrase. Which numbers exceed the human cross-lab variability band?
- (b) **(4 pts)** Classify the following claims as *SURVIVES* or *DOES NOT SURVIVE* this report, with one clause of justification each: (i) “silicon giving is positive and substantial”; (ii) “silicon mean giving is $\approx 30\%$ ”; (iii) “Model A exhibits human-like giving.”
- (c) **(4 pts)** Diagnose Model A’s Paraphrase 3: name two candidate explanations (one about the wording, one about the model) and the single follow-up run that would separate them.
- (d) **(3 pts)** State the “one model, one prompt” fallacy in one sentence, and the minimal robustness protocol your capstone pilot must therefore include.

Part B: Experimental Design (45 points)

Question 3 (The full sim-pilot workflow, pre-registered). A research team plans a human field experiment: does paying gig workers a fixed bonus upfront (versus the same bonus on completion) change task quality? The upfront bonus is unconditional — paid at the start and never clawed back. (Module 4’s gift exchange and Module 5’s reference dependence both make predictions; a clawback variant would be a different design, pitting loss framing against gain framing.) Before fielding it, they will run an LLM simulation pilot. Design the complete pilot workflow. Specify:

- (a) **(6 pts)** The human design being piloted: task, arms, outcomes (quantity AND quality), and the two theories’ competing predictions — so the pilot has something specific to serve.
- (b) **(12 pts)** The simulation pilot itself: agent design (personas, and what must NOT be smuggled into them), task rendering, treatments mirrored from the human design, and — the heart — the specific design parameters the pilot is allowed to tune (instruction wording, comprehension checks, bonus salience, quality-rubric clarity) versus the quantities it must not be used to estimate. Justify each assignment using the course’s can/cannot list.
- (c) **(9 pts)** The preregistration, written before any run: what gets committed (hypotheses, prompts, models, temperatures, N per cell, parsers, analysis code, the audit battery plan) and WHY cheap iteration makes this discipline more necessary, not less.
- (d) **(9 pts)** The decision rule: state, in advance, the three result patterns — one that justifies fielding the human study as designed, one that sends the team back to redesign (and what specifically gets redesigned), one that would halt the project — and defend why a silicon result can legitimately trigger each.
- (e) **(9 pts)** The steel-manned objection: write the *STRONGEST* version of “this pilot tells you nothing — the model has read your literature,” then the response that concedes what is correct and states precisely what the pilot still delivers.

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The committee member’s objection). At your capstone defense, a committee member says: “I stopped reading at the simulation section. These models are trained on the very literature you study; your ‘pilot’ is a book report with extra steps. Nothing in it should influence whether the human study runs.”

- (a) **(8 pts)** *Steel-man the objection: restate it in its STRONGEST form, adding the two best supporting arguments the committee member did not think to make (draw on the course's divergence taxonomy — contamination, alignment artifacts, missing channels, prompt fragility).*
- (b) **(9 pts)** *The defense: concede explicitly what is right (which uses of your pilot the objection genuinely kills), then defend what survives — the can-list uses — showing for each WHY the objection's mechanism does not touch it. Your defense must cite at least one concrete result from your own audit battery (real if you ran it; specified precisely if not).*
- (c) **(8 pts)** *Write the honest methods paragraph (4–6 sentences) for your capstone paper: what the simulation was used for, what it was not used for, the stability evidence, and where every effect-size and power assumption actually came from. This paragraph will be graded against the course's honesty standard: claiming a use from the cannot-list costs more than admitting a limitation.*

Attach your AI transcripts and, if you ran code, the results CSV.